

Exp No:4b

Data Visualization using Matplotlib

Date:

1. Line Chart

Aim

To visualize the variation in Views for different social media posts using a line chart in Matplotlib.

Algorithm

1. Import Pandas and Matplotlib.
2. Load the social media dataset.
3. Select Post_ID and Views.
4. Plot Post ID against Views using a line chart.
5. Add labels, title, and display the chart.

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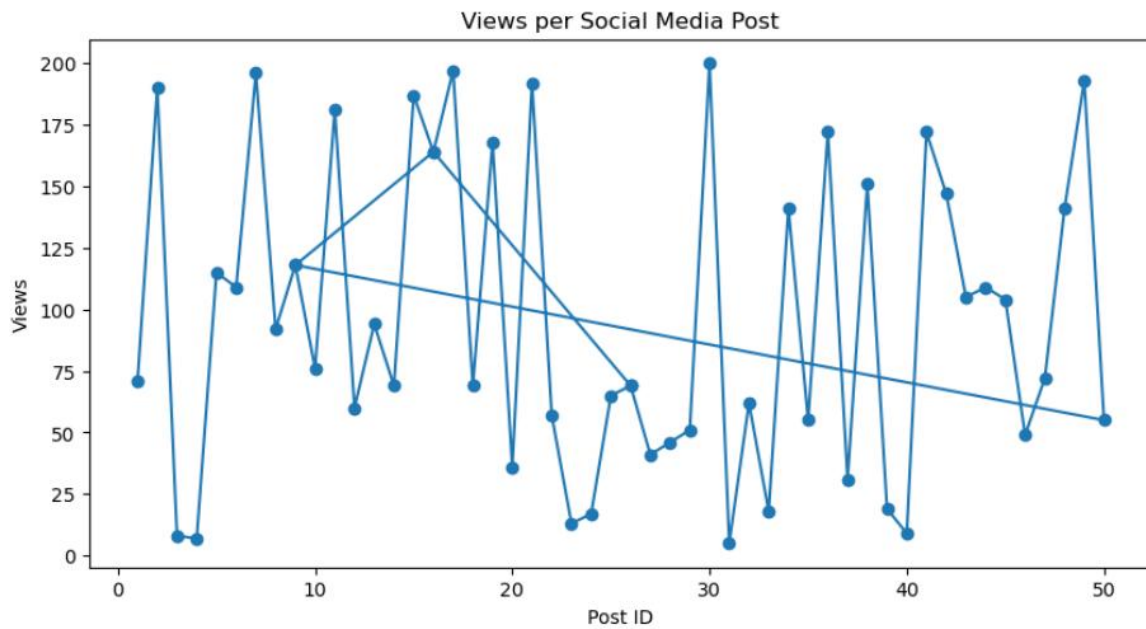
CODE:

```
import matplotlib.pyplot as plt

# Line chart
plt.figure(figsize=(10, 5))

plt.plot(df["Post_ID"], df["Shares"], marker="o")

plt.xlabel("Post ID")
plt.ylabel("Views")
plt.title("Views per Social Media Post")
plt.show()
```



2. Bar Chart

Aim

To compare the average number of Likes across different social media platforms using a bar chart.

Algorithm

1. Import Pandas and Matplotlib.
2. Load the social media dataset.
3. Calculate average Likes for each Platform.
4. Plot the values using a bar chart.
5. Add labels, title, and display the chart.

Code:

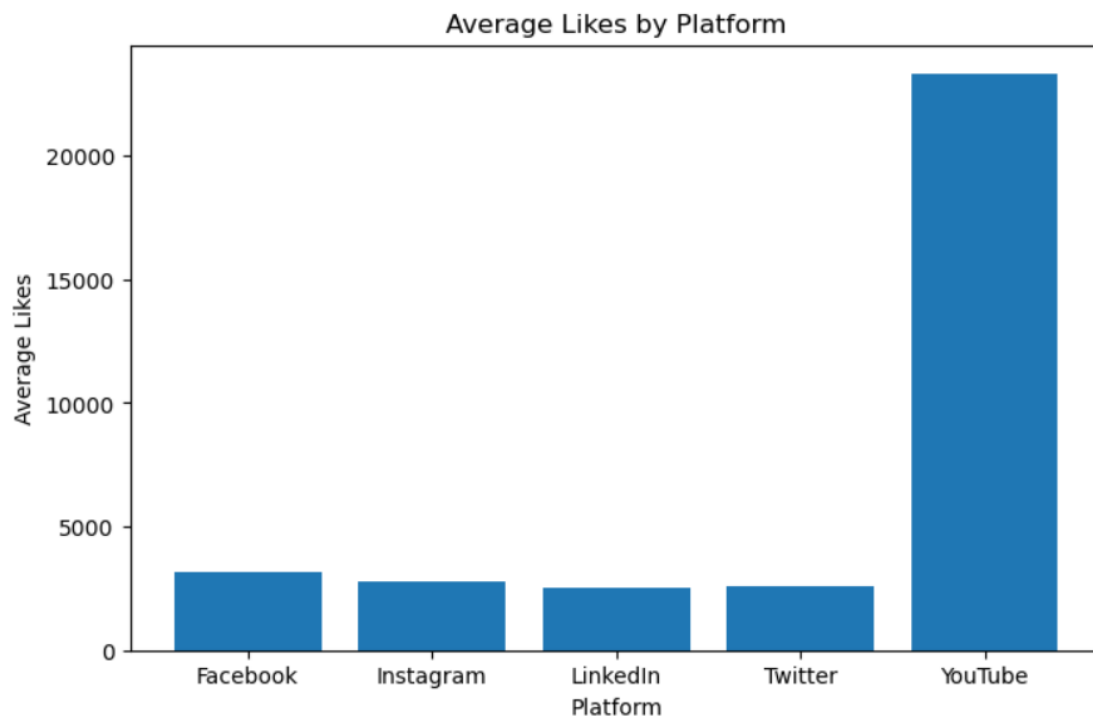
```
# Calculate average Likes by platform
platform_likes = df.groupby("Platform")["Likes"].mean()

# Bar chart
plt.figure(figsize=(8, 5))

plt.bar(platform_likes.index, platform_likes.values)

plt.xlabel("Platform")
plt.ylabel("Average Likes")
plt.title("Average Likes by Platform")

plt.show()
```



3. Pie Chart

Aim

To visualize the distribution of social media posts based on sentiment using a pie chart.

Algorithm

1. Import Pandas and Matplotlib.
2. Count the posts in each Sentiment category.
3. Plot the sentiment counts using a pie chart.
4. Add labels and percentages.
5. Display the chart.

Code:

```
# Count sentiment categories
sentiment_count = df["Sentiment"].value_counts()

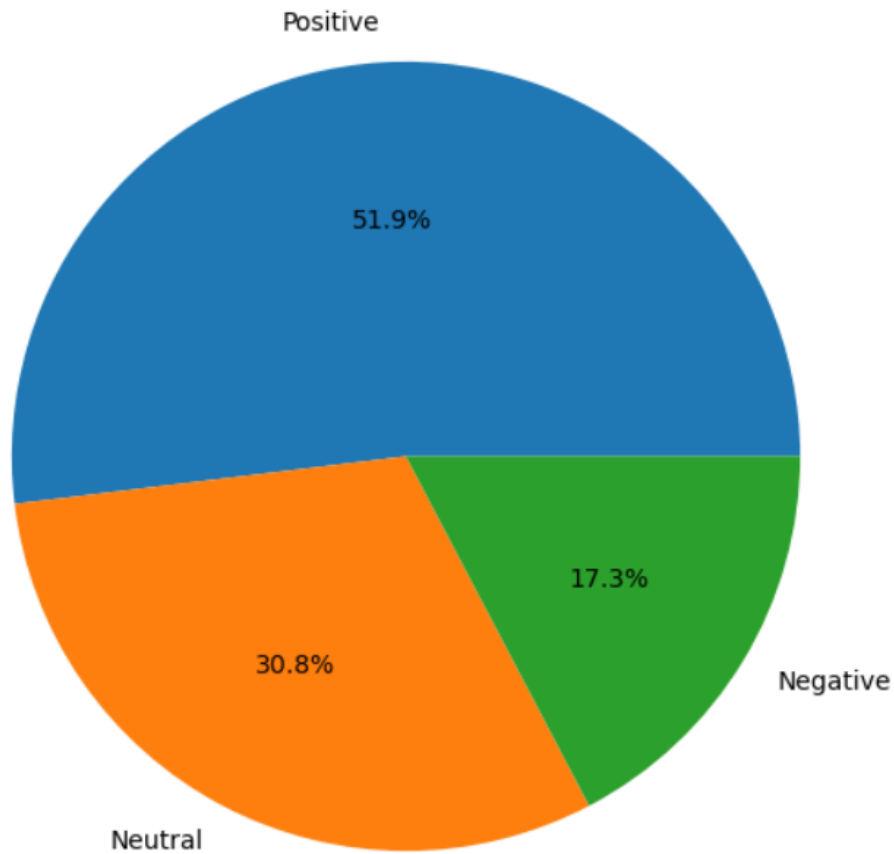
# Pie chart
plt.figure(figsize=(7, 7))

plt.pie(
    sentiment_count.values,
    labels=sentiment_count.index,
    autopct="%1.1f%%"
)

plt.title("Sentiment Distribution of Social Media Posts")

plt.show()
```

Sentiment Distribution of Social Media Posts



RESULT: Thus, the social media dataset was successfully visualized using Matplotlib through line, bar, and pie charts. The line chart showed the variation in views across posts, the bar chart compared average likes across platforms, and the pie chart represented the sentiment distribution of the posts.